

LONDON INTL, ON, Canada

WMO: 716230

Lat: 43.0269N

Lon: 81.1486W

Elev: 912

StdP: 14.22

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -2.2	(c) 3.0	(d) -9.7	(e) 3.4	(f) -1.2	(g) -3.8	(h) 4.6	(i) 4.0	(j) 28.5	(k) 22.9	(l) 26.1	(m) 22.9	(n) 7.2	(o) 0	(p) 0.560

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 18.5	(c) 86.5	(d) 72.2	(e) 83.8	(f) 71.0	(g) 81.2	(h) 69.3	(i) 74.9	(j) 82.5	(k) 73.1	(l) 80.5	(m) 71.5	(n) 78.4	(o) 10.3	(p) 200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.5	124.8	79.3	70.8	117.3	77.3	69.2	110.8	75.9	39.0	82.9	37.4	80.5	35.9	78.3	84.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.6	21.2	18.8	-9.1	91.0	6.1	3.4	-13.5	93.5	-17.1	95.4	-20.5	97.3	-25.0	99.8
			-9.6	78.9	6.1	2.2	-14.0	80.4	-17.6	81.7	-21.0	82.9	-25.4	84.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.0	22.2	23.7	32.3	44.5	56.4	66.1	69.9	68.3	61.9	50.3	38.7	28.7
	DBStd	18.95	11.41	10.71	10.86	8.71	8.31	6.64	5.40	5.08	6.92	8.11	8.54	9.17
	HDD50	3507	861	735	560	209	31	0	0	0	3	98	350	660
	HDD65	7018	1325	1155	1015	616	289	65	15	25	139	461	789	1124
	CDD50	2426	0	0	10	43	230	483	616	566	359	107	11	1
	CDD65	465	0	0	0	1	24	98	166	126	46	4	0	0
	CDH74	3562	0	0	2	13	224	775	1303	865	355	25	0	0
	CDH80	802	0	0	0	0	38	195	325	168	73	3	0	0
Wind	WSAvg	8.7	10.7	10.4	10.1	10.2	8.6	7.4	6.6	6.2	6.8	8.3	9.3	9.9
Precipitation	PrecAvg	40.0	3.5	2.6	3.0	3.1	3.5	3.5	3.5	3.0	3.6	3.4	3.5	3.5
	PrecMax	50.5	5.2	5.1	6.1	5.7	7.2	7.6	6.9	6.9	11.5	6.6	6.3	6.1
	PrecMin	29.0	2.0	0.7	1.2	1.2	0.7	1.7	0.7	0.5	0.7	0.9	1.1	1.7
	PrecStd	5.9	0.9	1.1	1.3	1.1	1.4	1.4	1.4	1.3	2.2	1.4	1.3	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.2	54.5	69.5	75.7	84.3	88.4	90.1	88.7	86.8	77.2	63.8	55.3
		MCWB	51.4	50.3	60.9	61.6	68.1	71.8	75.2	72.9	71.4	66.3	57.5	52.4
	2%	DB	47.2	46.5	59.6	69.4	80.0	85.2	86.7	84.2	81.6	71.7	59.2	49.4
		MCWB	45.8	43.3	51.6	57.4	66.7	71.2	72.7	72.1	69.3	63.1	54.1	47.1
	5%	DB	41.1	41.3	53.7	64.4	75.8	81.9	83.7	81.3	78.0	67.5	55.9	44.8
		MCWB	39.2	37.9	48.5	53.9	63.8	69.3	71.1	70.1	67.1	60.6	51.7	42.3
	10%	DB	36.9	37.2	47.8	59.5	71.3	78.4	80.8	78.9	74.5	63.5	52.1	40.3
		MCWB	35.0	34.7	42.8	50.3	60.4	67.0	69.3	68.8	65.5	58.0	48.1	38.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.5	51.6	61.3	63.6	71.7	75.5	78.1	77.0	74.9	68.9	59.1	52.5
		MCDB	52.9	53.9	69.6	72.1	81.2	84.7	86.9	82.6	82.7	74.9	62.3	54.3
	2%	WB	46.1	44.1	53.7	59.5	68.4	73.0	75.2	74.2	71.8	64.9	55.4	47.6
		MCDB	46.9	46.2	57.9	66.5	77.3	82.0	82.8	80.9	78.4	69.7	58.4	49.0
	5%	WB	39.6	38.4	48.6	56.0	65.7	71.1	73.3	72.3	69.4	61.5	52.0	42.5
		MCDB	41.0	40.8	53.4	62.4	73.1	79.1	80.5	78.6	74.9	66.2	55.3	44.3
	10%	WB	35.1	34.9	43.1	51.7	62.6	69.1	71.6	70.6	67.0	58.6	48.3	38.3
		MCDB	36.7	37.1	47.4	57.8	69.4	76.3	78.2	76.5	72.6	62.8	51.8	40.2
Mean Daily Temperature Range	5% DB	MDBR	12.6	13.9	15.0	18.1	19.1	18.3	18.5	18.3	19.1	16.1	12.9	10.7
		MCDBR	15.4	17.8	22.6	25.9	24.0	21.2	20.8	20.4	23.0	21.0	18.6	14.8
		MCWBR	15.1	15.7	16.8	17.1	14.0	11.2	10.7	10.9	12.6	13.4	14.8	13.4
	5% WB	MCDBR	15.0	17.3	21.4	22.8	21.3	18.5	17.6	17.3	19.0	18.8	17.0	14.4
		MCWBR	15.3	16.0	17.4	17.3	14.0	11.3	10.4	10.7	12.2	13.3	14.8	13.8
Clear-Sky Solar Irradiance	taub		0.293	0.327	0.332	0.369	0.395	0.426	0.415	0.399	0.373	0.357	0.315	0.289
	taud		2.391	2.262	2.384	2.316	2.284	2.223	2.278	2.304	2.371	2.400	2.482	2.457
	Ebn at Noon		268	273	289	285	279	269	271	272	271	260	258	259
	Edh at Noon		25	33	34	39	41	44	41	39	34	29	23	21
All-Sky Solar Radiation	RadAvg		456	674	1109	1441	1746	1954	1933	1692	1343	819	494	324
	RadStd		49	97	128	143	155	121	125	86	121	75	72	31

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (62 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page